

Vamshi Narayana Babu

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Software Engineer with 2+ years of experience building cloud-based applications, backend services, and distributed systems. Experienced in developing scalable software, and applying generative AI at the application level (OpenAI Agents SDK, AI-agent workflows). Strong in object-oriented design, REST API development, and CI/CD-driven operations for distributed systems.

EDUCATION

Arizona State University, Tempe, AZ

May 2026

Master of Science in Robotics and Autonomous Systems | **GPA: 3.8/4**

National Institute of Technology Karnataka, Surathkal, India

Jul 2018-May 2022

Bachelor of Technology in Electronics and Communication Engineering

TECHNICAL SKILLS

Languages: Python, Java, SQL.

Frontend: React.js, JavaScript, TypeScript, HTML5/CSS3.

Backend & Testing: FastAPI, Django, Java, Spring Boot, RESTful APIs, Jest, JUnit, Test-Driven Development.

Databases & Cloud: PostgreSQL, MongoDB, AWS (EC2, S3), CI/CD Pipelines, Docker, Kubernetes, Git.

Robotics: ROS2, SLAM, Perception, Path Planning, Motion Planning, Control Theory, Manipulation, HIL Testing.

Libraries: OpenCV, MoveIt, Nav2, RTABMap, Slam-Toolbox, YOLO, Camera SDKs, Pytorch, Scikit-learn, Pandas.

WORK EXPERIENCE

Stealth Startup, San Francisco, CA, USA

Jun 2025-Aug 2025

Software Engineer Intern

- Engineered a custom navigation stack for mobile robots using **C++**, **ROS2** and **LiDAR-SLAM**, implementing **optimized path-planning algorithms** to achieve 96% operational coverage.
- Refactored the computer vision pipeline with a **VLA**-based manipulation model (trained on 150+ datasets), improving grasp accuracy of the robotic arm **by 18%**.
- Redesigned the robotic arm training pipeline, transitioning from **imitation-only** learning to **reinforcement learning**, reducing training time **by 28%** and accelerating deployment of new manipulation behaviors.
- Integrated mobile robot's base and robotic arm into a unified pick-and-place system, utilizing **asynchronous communication** achieving **45+** picks/min across **40+** items with **83%** grasping accuracy.

Addverb Technologies, Noida, India

Aug 2022-Aug 2023

Software Engineer

- Architected and deployed a microservices-based warehouse automation system using **Java**, **Hibernate** and **Spring Boot**, supporting **55K+** daily orders across **4+** global production sites.
- Automated** inbound/outbound warehouse workflows, by developing a **data ingestion** service and **REST APIs**, integrating **TCP/IP** protocols with conveyor PLCs, achieving **80%** workflow automation.
- Built **event-driven** backend and **middleware** services using **Kafka** and **RabbitMQ** to process real-time telemetry and task coordination workflows, increasing operational throughput **by 50%**.
- Improved system reliability and observability by resolving **70+** production issues, implementing automated testing with **JUnit** and **Jest**, and developing **Grafana** dashboards to monitor service performance and stability.
- Containerized microservices with **Docker** and deployed them across **Kubernetes** environments, accelerating deployment workflows by **40%** while improving release consistency across development and production systems.

PROJECTS

Social Media Market Insights Agent | **Python, Playwright, ChromaDB, n8n**

- Designed an AI-powered data pipeline that collected social media content, stored trend embeddings, and leveraged summarization, sentiment analysis, and topic clustering to surface actionable market insights.
- Built automated workflows for data ingestion, categorization, and trend monitoring using agent-based processing pipelines.

Swipe-to-Shop & ML Recommendation Engine | **React, AWS(EC2, S3), FashionClip Model, Fast API**

- Architected a real-time clothing recommendation platform delivering personalized suggestions in under **200ms** by optimizing data flow between the AI model and cloud backend.

Multi-Robot Frontier Tracking & Exploration with ROS2 | **C++, MATLAB Simulink**

- Increased area exploration speed by **2.75×** by designing a distributed coordination framework for a multi robot fleet that uses path-planning and digital grid-mapping to assign unique goals and prevent teamwork conflicts.

Autonomous SLAM and Teleoperation with Turtlebot3 and ROS2 | **Python, SLAM, Gazebo, ROS2, Raspberry Pi**

- Engineered a real-time mapping and teleoperation stack, focusing on data processing and introspection that achieved **90%** autonomous coverage of the floor across both simulated and physical environments.

Bio-Inspired Grass Hopper Robot | **Mujoco, ESP32, Python, SG90 servos**

- Developed and validated a MuJoCo-based sim-to-real locomotion pipeline for a bio-inspired walking robot, achieving **5–22%** simulation-to-hardware error through system identification and gait optimization.